

EDUCATION

University of Michigan

Ann Arbor, MI

Master of Science in Mechanical Engineering (GPA 4.00/4.00, 1175±185hrs)

Anticipated May 2023

- Design Coursework: Design Optimization, Design of Experiment, Design with Material Selection, Design for Manufacturing, Mechatronics Design, Aircraft Design Optimization, Aerospace System Design, Graph Theory, Drawing
- Dynamic Coursework: Kinematics and Dynamics, Dynamics System Modeling, Robotic Dynamics, Robotic Manipulation, Differential Geometry, Numerical Analysis, Advanced Engineering Mathematics, Electromagnetism

Bachelor of Science in Aerospace Engineering (GPA 3.87/4.00, summa cum laude, 2336±348hrs) Sept. 2019 to May 2021

- Honor: James B Angell Scholar (2021), Dean's List (2021, 2020, 2019), University Honors (2020, 2019)

Shanghai Jiao Tong University (Dual Degree Program with University of Michigan)

Shanghai, China

Bachelor of Science in Mechanical Engineering (GPA 3.80/4.00, Ranking 2/55)

Sept. 2017 to Aug. 2021

- Honor: SJTU Excellent Engineering Undergrad Scholarship (3 out of 326, 2018)

EXPERIENCES

University of Michigan

Ann Arbor, MI

Design of Flapping-Wing Ornithopter with Prof. Sridhar Kota (2168±113hrs)

May 2020 to May 2022

- Designed, fabricated and tested my 1st ornithopter and all subsystems (drivetrain, linkage, embedded system, control, etc)
- Innovated a serial-elastic compound gear to reduce peak motor dynamics load and protect drivetrain from crash impact
- Modelled flight performance with bond-graph, Lagrangian dynamic, cycle-averaged aerodynamics, and aircraft dynamics
- Optimized the design by creating a gradient-based program after verifying the model with my previous flight test data

Flight Test of Retrospective Cost Adaptive Control with Prof. Dennis Bernstein's team (421±26hrs) Apr. 2022 to Present

- Designed, fabricated and tested a VTOL aircraft and a 5m-turning-radius aircraft for controller implementation
- Conducted physical flight tests and Gazebo/JSBSim PX4 Software-In-The-Loop simulation for fixed-wing and quadrotor

Design of Shape-Adapting Shell for Noninvasive Ventilator with Prof. Sridhar Kota (180±28hrs) Sept. 2022 to Present

- Designed, and simulated shape-adapting negative pressure ventilator with large-displacement compliant mechanisms
- Maximize the range of deformation by optimizing the sizing of compliant mechanisms based on FEA simulations

Design of Precision Binder Jet with Prof. Chinedum Okwudire (658±15hrs)

May 2022 to Sept. 2022

- Designed and simulated a 2μm-accuracy feed-drive platform for binder jet with deterministic design process
- Bridged connections with multiple companies to understand the industrial needs for binder jet process

Design of Air-puck and Hovercraft with Prof. Peter Washabaugh and team

Sept. 2019 to Dec. 2019

- Designed the shells and simulated inner and outer aerodynamics with Navier-Stokes equations using STAR-CCM+
- Fabricated the glass-fiber composition shells by designing and CNC routing a wax female mold with CATIA

Wild Independent Study of Mechanisms with Prof. Sridhar Kota (1149±486hrs)

Oct. 2019 to May 2022

- Learnt design of compliant mechanisms, design of nature, kinematical synthesis and analysis of advanced mechanisms

Shanghai Jiao Tong University (SJTU)

Shanghai, China

Design of Skiing-on-the-Sand Jumping Vehicle with Prof. Jaehyung Ju

May 2019 to Dec. 2019

- Designed, fabricated and tested a 120g vehicle with cam-follower jumping mechanism and sand-skiing origami

LEADERSHIP

University of Michigan

Ann Arbor, MI

Leader of SJTU Alumni Pickup Action (120±5hrs)

Aug. 2021 to Aug. 2021

- Initiated and managed a 16-people team to pick up over 200 SJTU students arriving at DTW airport from over 40 flights

Shanghai Jiao Tong University

Shanghai, China

President of Libraries and Museums Volunteer Association

Oct. 2018 to Aug. 2019

- Managed volunteer activities of 900+ students and 7000+ person-hours at 6 libraries and museums

Hours were recorded between Sept. 1 2019 and Nov. 2 2022 with Google Calendar, and were counted with three different methods using my customized code